

Bias-Variance Tradeoff

The Fundamental Tradeoff in Machine Learning

Total Error = Bias² + Variance + Irreducible Error

- Bias: Error from wrong assumptions
- Variance: Error from sensitivity to training data
- Goal: Minimize both!

```
In [1]: import numpy as np
import matplotlib.pyplot as plt
from matplotlib.patches import Circle
import seaborn as sns

plt.style.use('dark_background')
sns.set_palette("bright")
np.random.seed(42)
```

1. The Dartboard Analogy

Classic visualization of bias and variance

```
In [2]: # Create dartboard visualizations
fig, axes = plt.subplots(2, 2, figsize=(14, 14))

scenarios = [
    {'bias': 'low', 'variance': 'low', 'title': 'Low Bias, Low Variance\n(IDEAL)'},
    {'bias': 'low', 'variance': 'high', 'title': 'Low Bias, High Variance\n(PRECISE but INACCURATE)'},
    {'bias': 'high', 'variance': 'low', 'title': 'High Bias, Low Variance\n(BUT INACCURATE)'},
    {'bias': 'high', 'variance': 'high', 'title': 'High Bias, High Variance\n(WRONG and UNRELIABLE)'}
]

for ax, scenario in zip(axes.flat, scenarios):
    # Draw dartboard circles
    for radius in [1, 2, 3, 4]:
        circle = Circle((0, 0), radius, fill=False, edgecolor='white',
                        linewidth=2, alpha=0.3)
        ax.add_patch(circle)

    # Draw center (bullseye)
    ax.scatter(0, 0, s=300, c='yellow', marker='*',
              edgecolors='white', linewidth=2, zorder=5, label='Target')

    # Generate dart throws based on bias and variance
    n_darts = 20

    if scenario['bias'] == 'low':
        center_x, center_y = 0, 0 # Centered on target
```

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else:
    center_x, center_y = 2.5, 2.5 # Off-center

if scenario['variance'] == 'low':
    spread = 0.3 # Tight cluster
else:
    spread = 1.2 # Scattered

# Generate dart positions
darts_x = np.random.normal(center_x, spread, n_darts)
darts_y = np.random.normal(center_y, spread, n_darts)

# Plot darts
ax.scatter(darts_x, darts_y, s=150, c=scenario['color'],
           edgecolors='white', linewidth=2, alpha=0.8, label='Predictions')

# Draw mean prediction
mean_x, mean_y = np.mean(darts_x), np.mean(darts_y)
ax.scatter(mean_x, mean_y, s=400, c='cyan', marker='X',
           edgecolors='white', linewidth=3, zorder=4, label='Average Prediction')

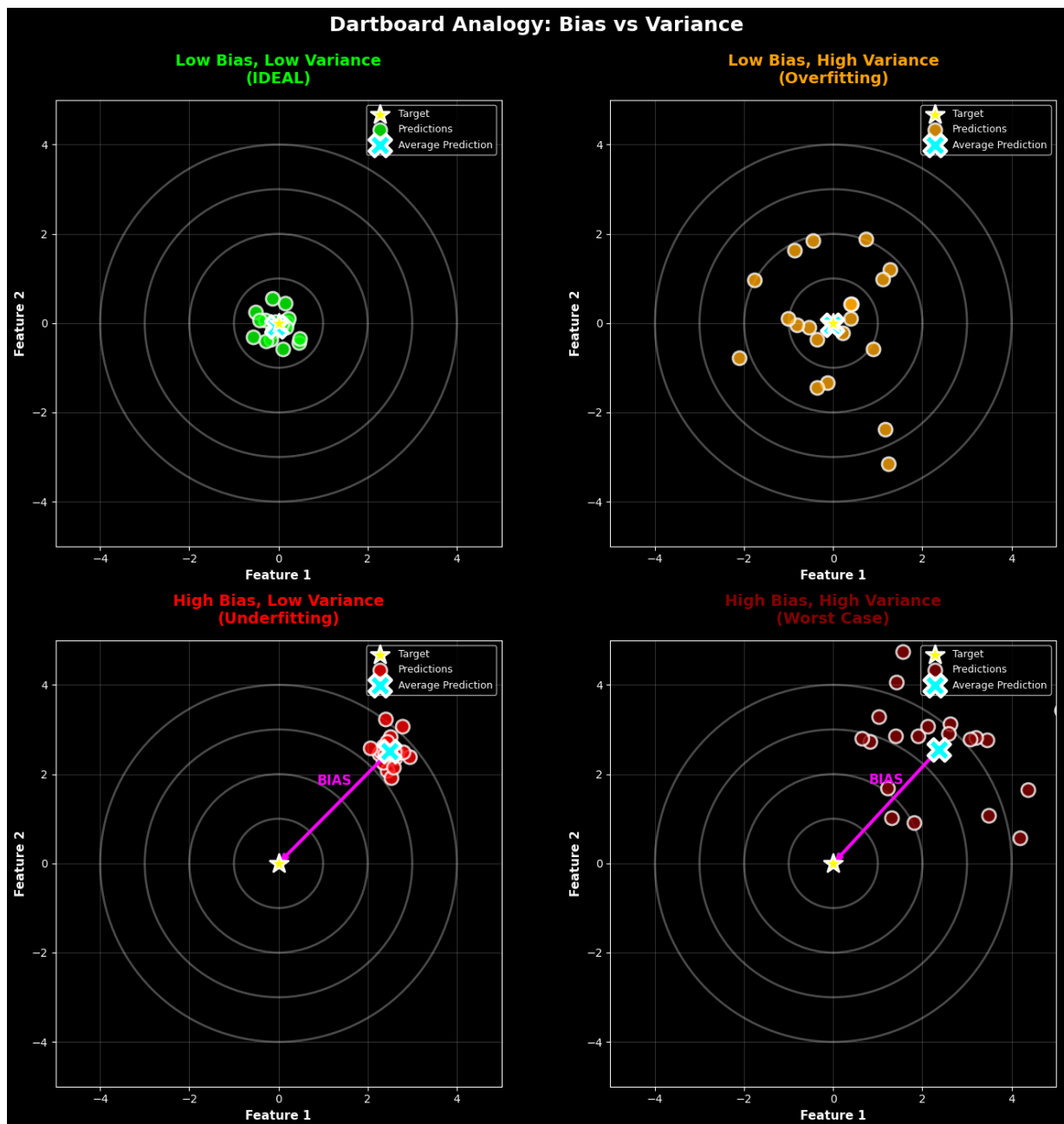
# Draw bias arrow (from target to average)
if scenario['bias'] == 'high':
    ax.annotate('', xy=(mean_x, mean_y), xytext=(0, 0),
               arrowprops=dict(arrowstyle='<->', color='magenta', lw=3))
    ax.text(mean_x/2, mean_y/2 + 0.5, 'BIAS', fontsize=12,
           fontweight='bold', color='magenta', ha='center')

ax.set_xlim(-5, 5)
ax.set_ylim(-5, 5)
ax.set_aspect('equal')
ax.set_title(scenario['title'], fontsize=14, fontweight='bold',
            color=scenario['color'], pad=15)
ax.legend(fontsize=9, loc='upper right')
ax.grid(alpha=0.2)
ax.set_xlabel('Feature 1', fontsize=11, fontweight='bold')
ax.set_ylabel('Feature 2', fontsize=11, fontweight='bold')

plt.suptitle('Dartboard Analogy: Bias vs Variance', fontsize=18, fontweight='bold')
plt.tight_layout()
plt.show()

print("Dartboard Interpretation:")
print("="*60)
print("Bullseye (yellow star) = True target")
print("Darts (colored dots) = Individual predictions")
print("Cyan X = Average prediction\n")
print("BIAS = Distance from average prediction to target")
print("VARIANCE = Spread of predictions around average\n")
print("IDEAL: Tight cluster around bullseye (low bias, low variance)")
print("="*60)

```



Dartboard Interpretation:

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Bullseye (yellow star) = True target

Darts (colored dots) = Individual predictions

Cyan X = Average prediction

BIAS = Distance from average prediction to target

VARIANCE = Spread of predictions around average

IDEAL: Tight cluster around bullseye (low bias, low variance)

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2. Mathematical Decomposition

$$\text{Error} = \text{Bias}^2 + \text{Variance} + \text{Noise}$$

```
In [3]: # Simulate bias and variance for different model complexities
model_complexity = np.linspace(1, 20, 50)

# Bias decreases with complexity (exponential decay)
bias_squared = 5 * np.exp(-0.3 * model_complexity) + 0.1

# Variance increases with complexity (exponential growth)
variance = 0.05 + 0.01 * (model_complexity ** 1.5)

# Irreducible error (constant)
irreducible_error = np.ones_like(model_complexity) * 0.5

# Total error
total_error = bias_squared + variance + irreducible_error

# Find optimal complexity
optimal_idx = np.argmin(total_error)
optimal_complexity = model_complexity[optimal_idx]

print(f"Optimal Model Complexity: {optimal_complexity:.2f}")
print(f"Minimum Total Error: {total_error[optimal_idx]:.3f}")
print(f" - Bias² at optimal: {bias_squared[optimal_idx]:.3f}")
print(f" - Variance at optimal: {variance[optimal_idx]:.3f}")
print(f" - Irreducible error: {irreducible_error[optimal_idx]:.3f}")
```

```
Optimal Model Complexity: 11.47
Minimum Total Error: 1.199
 - Bias² at optimal: 0.260
 - Variance at optimal: 0.438
 - Irreducible error: 0.500
```

3. The Tradeoff Curve

The classic U-shaped total error

```
In [4]: fig, ax = plt.subplots(figsize=(14, 8))

# Plot individual components
ax.plot(model_complexity, bias_squared, 'red', linewidth=4,
        label='Bias² (Model Error)', alpha=0.8)
ax.plot(model_complexity, variance, 'orange', linewidth=4,
        label='Variance (Model Sensitivity)', alpha=0.8)
ax.plot(model_complexity, irreducible_error, 'gray', linewidth=3,
        linestyle='--', label='Irreducible Error (Noise)', alpha=0.6)

# Plot total error
ax.plot(model_complexity, total_error, 'cyan', linewidth=5,
        label='Total Error', alpha=0.9)

# Mark optimal point
ax.scatter(optimal_complexity, total_error[optimal_idx],
          s=600, c='lime', marker='*', edgecolors='yellow',
          linewidth=3, zorder=5, label='Optimal Complexity')
```

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# Shade regions
ax.axvspan(1, 8, alpha=0.15, color='red', label='Underfitting Zone')
ax.axvspan(8, 12, alpha=0.15, color='lime', label='Sweet Spot')
ax.axvspan(12, 20, alpha=0.15, color='orange', label='Overfitting Zone')

# Add annotations
ax.annotate('High Bias\nSimple Model', xy=(3, 4), fontsize=13,
            fontweight='bold', color='red', ha='center',
            bbox=dict(boxstyle='round', facecolor='black', alpha=0.7))

ax.annotate('Balanced\nTRADE-OFF', xy=(optimal_complexity, total_error[optimal_complexity]),
            fontsize=13, fontweight='bold', color='lime', ha='center',
            bbox=dict(boxstyle='round', facecolor='black', alpha=0.7),
            arrowprops=dict(arrowstyle='->', color='lime', lw=2))

ax.annotate('High Variance\nComplex Model', xy=(17, 6), fontsize=13,
            fontweight='bold', color='orange', ha='center',
            bbox=dict(boxstyle='round', facecolor='black', alpha=0.7))

# Arrow showing tradeoff
ax.annotate('', xy=(15, bias_squared[40]), xytext=(15, 0.5),
            arrowprops=dict(arrowstyle='<->', color='red', lw=2))
ax.text(15.5, 1.5, 'Bias\nDecreases', fontsize=11, color='red', fontweight='bold')

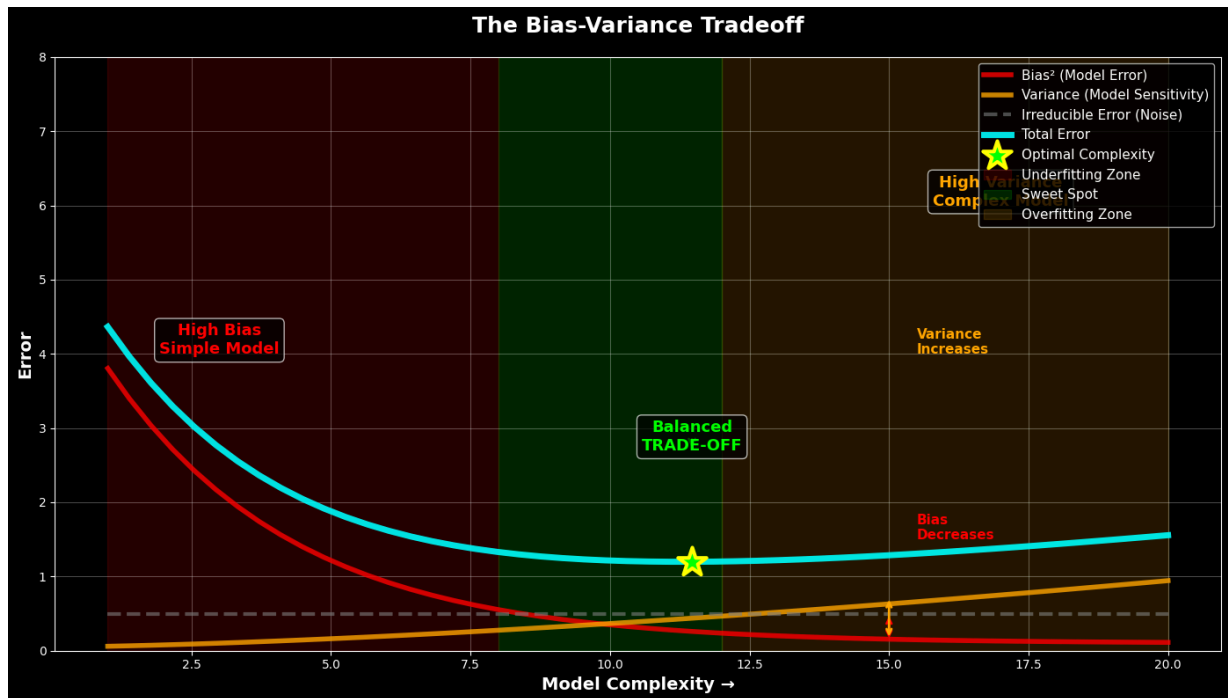
ax.annotate('', xy=(15, variance[10]), xytext=(15, variance[40]),
            arrowprops=dict(arrowstyle='<->', color='orange', lw=2))
ax.text(15.5, 4, 'Variance\nIncreases', fontsize=11, color='orange', fontweight='bold')

ax.set_xlabel('Model Complexity →', fontsize=14, fontweight='bold')
ax.set_ylabel('Error', fontsize=14, fontweight='bold')
ax.set_title('The Bias-Variance Tradeoff', fontsize=18, fontweight='bold', pad=10)
ax.legend(fontsize=11, loc='upper right')
ax.grid(alpha=0.3)
ax.set_ylim([0, 8])

plt.tight_layout()
plt.show()

print("\nKey Insights:")
print("="*60)
print("As Model Complexity INCREASES:")
print("  - Bias DECREASES (better fit to training data)")
print("  - Variance INCREASES (more sensitive to data)")
print("\nOptimal Model: Minimizes TOTAL error (bias² + variance)")
print("\nU-shaped curve: Too simple OR too complex = HIGH error!")
print("="*60)

```



Key Insights:

As Model Complexity INCREASES:

- Bias DECREASES (better fit to training data)
- Variance INCREASES (more sensitive to data)

Optimal Model: Minimizes TOTAL error ($\text{bias}^2 + \text{variance}$)

U-shaped curve: Too simple OR too complex = HIGH error!

4. Visual Summary

Quick reference comparison

```
In [5]: # Create comparison table
fig, ax = plt.subplots(figsize=(14, 6))

# Hide axes
ax.axis('off')

# Table data
table_data = [
    ['Characteristic', 'HIGH BIAS\n(Underfitting)', 'BALANCED\n(Optimal)', 'HIGH VARIANCE\n(Overfitting)'],
    ['Model Complexity', 'Too Simple', 'Just Right', 'Too Complex'],
    ['Training Error', 'High', 'Low', 'Very Low'],
    ['Test Error', 'High', 'Low', 'High'],
    ['Train-Test Gap', 'Small', 'Small', 'Large'],
    ['Generalizes?', 'x No', '✓ Yes', 'x No'],
    ['Problem', 'Oversimplified', '-', 'Memorized'],
    ['Solution', 'Increase complexity', '-', 'Reduce complexity\nGet more data']
]

colors = [['white', 'red', 'lime', 'orange']] + \
```

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[['lightgray', '#4d0000', '#004d00', '#4d2600']] * (len(table_data))

table = ax.table(cellText=table_data, cellLoc='center', loc='center',
                 cellColours=colors, colWidths=[0.25, 0.25, 0.25, 0.25])

table.auto_set_font_size(False)
table.set_fontsize(11)
table.scale(1, 3)

# Style header row
for i in range(4):
    cell = table[(0, i)]
    cell.set_text_props(weight='bold', size=12)
    cell.set_facecolor('darkgray')

# Style first column
for i in range(len(table_data)):
    cell = table[(i, 0)]
    cell.set_text_props(weight='bold')

plt.title('Bias vs Variance: Quick Reference', fontsize=16, fontweight='bold')
plt.tight_layout()
plt.show()

```

Bias vs Variance: Quick Reference			
Characteristic	HIGH BIAS (Underfitting)	BALANCED (Optimal)	HIGH VARIANCE (Overfitting)
Model Complexity	Too Simple	Just Right	Too Complex
Training Error	High	Low	Very Low
Test Error	High	Low	High
Train-Test Gap	Small	Small	Large
Generalizes?	✗ No	✓ Yes	✗ No
Problem	Oversimplified	—	Memorized
Solution	Increase complexity	—	Reduce complexity Get more data

5. Key Takeaways

The Fundamental Equation:

Expected Test Error = Bias² + Variance + Irreducible Error

Definitions:

Bias (Systematic Error):

- Error from wrong assumptions in the model
- High bias = Model too simple (underfitting)
- Can't capture underlying pattern

- Training error is high

Variance (Random Error):

- Error from sensitivity to training data
- High variance = Model too complex (overfitting)
- Captures noise as if it's pattern
- Training error is low, test error is high

Irreducible Error:

- Inherent noise in data
- Cannot be reduced by any model
- Lower bound on achievable error

The Tradeoff:

- **Decrease bias** → Usually increases variance
- **Decrease variance** → Usually increases bias
- **Goal:** Find sweet spot that minimizes total error

How to Balance:

Reduce Bias:

- Use more complex model
- Add more features
- Train longer
- Reduce regularization

Reduce Variance:

- Get more training data (best!)
- Simplify model
- Use regularization (L1/L2)
- Ensemble methods (bagging)
- Cross-validation
- Dropout

Diagnostic:

- **High training error** → High bias (underfit)
- **Large train-test gap** → High variance (overfit)
- **Both low** → Optimal! ✓

Remember:

You can't eliminate both bias and variance completely!

The art of ML = Managing this tradeoff! 🎯